SAMPLE

Famous Astronomers: Ruby Payne-Scott

Radio Astronomy Trailblazer

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Free sample: Introduction & Chapter 1

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Introduction

In the spring of 1945, while the Second World War was still burning in the Pacific, a physicist at a radar station outside Sydney turned an antenna toward the rising Sun and heard static spike with the sunrise. Her name was Ruby Payne-Scott, and that afternoon's work, improvised on wartime instruments in the narrow time the equipment could be spared from aircraft tracking, opened the study of the Sun at radio wavelengths and made her one of the world's first radio astronomers, in a field that did not yet have the name. Within six years she had found that the Sun emits bursts of radio noise tied to its storms, measured the million-degree atmosphere those bursts traverse, and helped invent the technique that all modern radio telescopes rest on. Then the policy

of her own government ended her career, and for half a century her name lived mainly in footnotes.

This book tells the whole story: the physicist, the discoveries, the instrument, the injustice, and the rediscovery. Ruby Payne-Scott's scientific career fits into a single decade, 1941 to 1951, and almost all the lasting work into the last six years of it, which makes her one of the most compact careers in the history of astronomy; but the decade's work seeded a century. The burst types her group identified are still the standard categories of solar radio emission, the basis of space-weather forecasting; the interferometer she helped build, using the ocean as a mirror, was the first step on the road to the instruments that now map the sky in radio waves; and the fight she lost, the forced resignation of a married woman, became decades later a landmark in the history of women in science, a case study of talent discarded by policy.

Her life also opens a window on a remarkable institution at a remarkable moment. The Radiophysics Laboratory of the Australian scientific research service, built to develop radar for the Pacific war, found itself at war's end with a stock of surplus antennas, receivers, and trained young physicists, and its chief, with a stroke of organizational genius, aimed the whole apparatus at the sky. Australia became, almost overnight, one of the two or three world centers of a brand-new science, and Payne-Scott was one of its founders: the best mathematician in the group, her colleagues said, a gifted physicist, and a woman in an institution whose rules were written as if she did not exist.

The chapters that follow keep her at the center but give the science its full weight, because the science is what was nearly lost. The burst taxonomy, the hot corona, the Fourier trick that turns a coastline into a telescope: these are not biographical color, they are the foundations of heliophysics and radio astronomy as practiced today. The book ends with the long afterlife, the medal and the lecture that carry her name, the ferry on Sydney Harbour, and the

historians who pieced her career back together, because the story of Ruby Payne-Scott is finally a story about what science owes to the people it forgets, and about how a field, late and imperfectly, pays its debts.

Chapter 1: The Discovery of Ruby Payne-Scott

Ruby Payne-Scott's life has two discovery stories: her own discoveries, and the world's discovery of her.

She was born Ruby Violet Payne-Scott on 28 May 1912, in Grafton, New South Wales, a river town in the state's north, where her father was an accountant; the family, the surname was already hyphenated, and it was her birth name, not a marriage, moved to Sydney, and she grew up a city girl with a country

birth certificate. She was a gifted student in an era when gifted girls were steered firmly toward teaching, and she steered herself instead into physics: a bachelor of science at the University of Sydney completed in 1933, a master's in 1936, with research on the cyclotron, the particle accelerator then at the cutting edge of nuclear physics.

The physics department at Sydney was one of the few places in the country where a woman could do serious research, and Payne-Scott, working under the physicist who later led the university's war effort, made the most of it. The university years deserve their context, because the context is part of the story. In the Australia of the early 1930s, a woman doing an honors physics degree was a rarity her lecturers could count on one hand; the profession's standard route for a gifted woman was schoolteaching, and the country's entire research establishment was small enough that a science career, for anyone, was an act of persistence.

Payne-Scott's generation of Sydney physics students, taught in a department with real research ambitions and wartime ties forming, would supply a surprising share of the country's post-war scientific leadership, and the women in that generation, few as they were, had to be twice as good to stay; by every colleague's testimony, she met the standard without apparent effort. Her first paid science, though, was medical: she spent the late 1930s as a physicist with the Cancer Research Committee at the University of Sydney, working with radium and X-rays on the treatment of tumors, a field where a physicist's skill at radiation dosimetry was directly lifesaving. The radium work deserves its own lines, because it was real physics with real stakes. The Cancer Research Committee's program, the treatment of tumors by

implanted radium needles and directed X-rays, needed a physicist's precision, the placement of sources, the calculation of doses, the measurement of the radiation actually delivered, and the young Payne-Scott's contributions, by the

era's publications and her supervisor's testimony, were substantial enough that her wartime hiring was eased by a reputation for careful measurement already in place. The medical apprenticeship also taught the discipline of dangerous equipment: radium and high voltage demand procedures, logs, and respect, and the habits she formed in the cancer ward served the radar bench and then the antenna sites, the same craft applied to different radiations. She was, by the outbreak of war, one of a very small number of women in Australia qualified as a research physicist, and the war, which drained the country's scientific manpower, would open the door that peacetime had kept shut.

The door was radar. In 1941 the Radiophysics Laboratory of the Council for Scientific and Industrial Research, the CSIR, the ancestor of today's CSIRO, was

set up at the University of Sydney to develop radar for the Pacific theater. The laboratory's founding is a story in itself: the British radar mission of 1939-1940, which carried the cavity magnetron's secrets across the world, seeded radar establishments throughout the Commonwealth, and Australia's, assembled at the University of Sydney and operational by 1941, grew into one of the largest scientific organizations the country had fielded, its staff in the hundreds, its stations strung along the coast, its workshops and test gear a small industrial empire, and the manpower shortage meant that women were hired for scientific posts at all.

Payne-Scott joined as a research officer, one of the first women on the scientific staff, and spent the war years on radar's hardest practical problems: the detection of aircraft off the coast, the behavior of echoes over water, the effects of the sea surface on radar returns. The work was classified, collaborative, and instrument-rich, and it taught her the radio craft that her post-war career would spend: how to build receivers sensitive enough to hear the sky, how to calibrate against a known noise source, and how to push an antenna and a receiver, night after night, to their physical limits. The laboratory's wartime culture, egalitarian, urgent, and young, was the incubator of everything that followed.

Her own discoveries came fast at the war's end. The radar chain had been built to detect the Sun, in effect, the operators knew the Sun was a radio noise source, because it jammed their sets at sunrise, and in 1945, with the war over and the Pacific still in the last days of its fighting, the laboratory's leadership decided to investigate the phenomenon before the equipment was demobilized. Payne-Scott, with her colleagues Joseph Pawsey and Lindsay McCready among others, carried out the first Australian radio observations of the Sun that spring: October 1945, at the Collaroy and Dover Heights radar stations on the coast near Sydney, instruments borrowed between air-defense

shifts, a genuine after-hours science.

The results were immediately strange and immediately publishable: the Sun's radio emission was vastly stronger than any thermal calculation allowed, and it varied, day to day, in ways that tracked the sunspots. The era's explanation, that something on the Sun was hotter than physics permitted, would bend over the next two years into the discoveries that made the field: the corona is a million degrees hot, and its storms emit bursts of radio noise that a physicist can classify, time, and explain.

The world's discovery of her took longer, and it is a story of archival recovery. Payne-Scott left science in 1951, at thirty-nine, when her employer learned what the Commonwealth's rules required it to learn, that she was a married woman, and she spent the rest of her working life as a mathematics teacher

and her retirement as a mother and bushwalker, seldom speaking of her research years. The historical record survived in the papers, the classified reports declassified, the correspondence preserved, and in the footnotes of the field's histories, which credited the group's work while often omitting the woman who did the mathematics.

The recovery began in earnest in the 1990s and 2000s, driven by historians of Australian science and by radio astronomers who had come to understand how much of their technique's pedigree ran through her hands, and it culminated in a full-scale biography published as part of the International Year of Astronomy's She is an Astronomer program, and in the honors that followed, a career medal, a namesake lecture, and eventually a harbour ferry, that Australia now attaches to her name. Both discovery stories, hers and ours, are told in the chapters that follow.

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